Project Workflow

The following detailed workflow outlines the complete transaction lifecycle of **HarvestHub**. It enforces a strict "Payment-First" policy, ensuring the Farmer only begins harvesting after the Buyer has financially committed.

Phase 1: Market Analysis & Trusted Listing Creation

- **Step 1.1: Market Insight Analysis**
    - **Action:** The Farmer accesses the **Market Insights Dashboard** to analyze current demand trends (e.g., "High demand for Big Onions in Dambulla").
    - **System Status:** N/A (Informational View)
- **Step 1.2: Location-Locked Initiation**
    - **Action:** The Farmer initiates a **Supply Request**. The system performs a background **GPS Location Check**.
    - **Constraint:** The Farmer must be physically present at their registered farm address to proceed.
    - **System Status:** Drafting

Step 1.3: Smart Guided Capture

- **Action:** The system activates the Real-Time Camera (Gallery upload is disabled). The app displays on-screen guides prompting the Farmer to capture:
    - **Image 1 (Wide Angle):** To show the cultivated field area and crop density for estimating the overall harvest volume.
    - **Image 2 (Close Up):** To show the texture, size, and expected quality grade of a freshly harvested sample from the field.
    - **Video Clip (10s):** To demonstrate the overall field condition alongside a 360-degree view of the harvested sample to prove authenticity and freshness.
- **Note:** All media is automatically geotagged with GPS coordinates.
- **System Status:** Drafting

***Step 1.4: Submission**

- **Action:** The Farmer enters the expected price per kg and the **Estimated Total Yield (in KG)** based on their field size and historical data, then submits the listing.
- **Constraint:** The system explicitly labels this as an "Estimated Quantity" on the buyer's feed, allowing for a standard real-world variance (e.g., +/- 5%) when the final physical weighing occurs at the loading phase.
- **System Status:** Pending Verification

Step 1.5: Collaborative Officer Quality Check & Queueing

- **Action (Queue Entry):** The submitted listing enters a global "Pending Verification Queue" that is visible to all registered Agricultural Officers on their JavaFX dashboard.
- **Action (Claiming):** An available Agricultural Officer selects a listing from the queue. The system immediately temporarily locks this listing (Status: *Under Review*) so that no other officer can grade it at the same time.
- **Action (Grading):** The officer reviews the GPS data and the Smart Guided photos. They have three options:
    1. **Approve & Grade:** Assign a quality grade (e.g., Grade A, B, C).
    2. **Request Retake:** Ask the farmer to upload a clearer photo (Status returns to *Drafting*).
    3. **Reject:** Reject the listing if it violates platform guidelines.
- **System Status:** Listed (if approved) / Action Required (if retake/rejected).

Phase 2: Discovery, Negotiation & Scheduling

- **Step 2.1: Initial Engagement (Bidirectional)**

- **Action (Scenario A - Supply Driven):** A Buyer browses the verified marketplace listings and sends a "Purchase Request" to a Farmer.
- **Action (Scenario B - Demand Driven):** A Farmer browses the Buyer "Wanted Ads" and sends a "Supply Offer" to a Buyer, detailing their estimated yield and expected price.
- **System Status:** Negotiating

- **Step 2.2: Negotiation & Scheduling**
  - **Action:** Following the initial request or offer, the Farmer and Buyer use the in-app chat to finalize the exact quantity, agree on a final price per kg, and establish a specific Pickup Date.
  - **System Status:** Accepted (Order is locked, but not paid).

Phase 3: Automated Logistics Dispatch & Financial Commitment

- **Step 3.1: Automated Vehicle Selection & Costing**
  - **Action:** Once the Buyer and Farmer confirm the order, the system's Logistics Engine reads the total harvest weight.
  - **Action:** The system eliminates manual selection by automatically filtering and assigning the most appropriate vehicle category based on weight capacity.
  - **Action:** The system fetches GPS coordinates via the Google Maps API and automatically calculates a dynamic transport fee Cost = (Distance × Rate) + (Weight × HandlingFee).
  - **System Status:** Awaiting Payment

Step 3.2: Secure Advance Payment

- **Action:** The system calculates the total bill (Total Transport Fee + Partial Crop Cost) and automatically adds a **3% Buyer Convenience Fee**.

- **Action:** The Buyer pays this total Advance Payment through the system’s secure gateway into the platform's central holding account.

- **System Status:** Payment Processing / Dispatching

- **Step 3.3: Smart Dispatch & Harvest Trigger**
    - **Trigger:** Successful Advance Payment verification.
    - **Action (Dispatch Loop):** The system automatically sends a transport request to the nearest available matched Transporter.
    - **Scenario A (Accepted):** The Transporter accepts. The system sends a "Vehicle Assigned - Start Harvesting" push notification to the Farmer.
    - **Scenario B (Declined/No Vehicles):** The system intelligently reroutes the request to the next available driver, or pauses until a vehicle is online.
    - **Constraint:** The Farmer must not start harvesting until the specific notification is received.

Phase 4: Execution (Harvest, Proof-of-Loading & Transit)

- **Step 4.1: Harvest Completion & Notification Action:** The Farmer receives the "Start Harvesting" notification, harvests the crops, and packs them.
    - **Action:** The Farmer taps "Harvest Ready" on the app.
    - **Trigger:** The system pings the assigned Transporter with the exact GPS pickup coordinates.
    - **System Status:** Ready for Pickup

Step 4.2: Pre-Loading Quality Verification & Liability Checkpoint

- **Action (Mandatory Evidence):** Upon arrival at the farm, the transport Helper physically inspects the harvest and captures time-stamped, geotagged photographs of the crops *before* loading them into the vehicle. The Transporter App locks the "Start Journey" button until these photos are uploaded.

- **Scenario A (Approved):** The Buyer remotely reviews and approves the photos via their app. The Helper proceeds to load the goods into the vehicle, and the Transporter is allowed to press "Start Journey".

- **Scenario B (Pre-Loading Rejection):** If the Buyer spots poor quality or incorrect crops in the photos, they hit "Reject." The loading process is aborted. The Buyer receives a 100% refund of their amount, and the Farmer's wallet is automatically penalized with an "Empty Trip Fee" to compensate the transport team for the wasted trip.

- **System Status:** Verified & Loaded (If Approved)

Step 4.3: Automated Payment Release & Live Tracking

- **Trigger:** The Transporter updates the app status to "Start Journey".
- **Action (Payment Release):** The system automatically transfers the Advance Payment (Crop Portion) directly into the Farmer's wallet. This guarantees the Farmer gets paid the exact second the goods leave their land.
- **Action (Live Tracking):** The Transporter app begins sending continuous GPS location updates to the Buyer's screen so they can track the delivery in real-time.
- **System Status:** In Transit (Live Tracking Active)

Phase 5: Delivery, Rejection & Automated Final Settlement

- **Step 5.1: Physical Arrival & Quality Check**
    - **Action:** The vehicle arrives. The Buyer physically inspects the goods against the pre-loading photos.
    - **Scenario A (Accepted):** The Buyer approves the quality and pays the remaining 80% balance through the system.
    - **Scenario B (Post-Delivery Rejection):** The Buyer discovers transit damage and hits "Reject Delivery" (Triggering Step 5.3).
- **Step 5.2: Unloading & 2-Way Final Confirmation (If Accepted)**

 Action: The Helper unloads the goods.

 Action (Verification): To prevent delivery fraud, the system requires a digital confirmation from both parties present:

- **Driver:** Taps "Delivery Completed" on their app at the drop-off location.
- **Buyer:** Taps "Goods Received" on their app.

 Action (Automated Settlement): Once the system receives both confirmations, it acts as the central accountant:

1. **Platform Revenue:** The system retains the 3% Buyer Convenience Fee.
2. **Farmer Payment:** The system transfers the final crop payment to the Farmer (with a 0% platform fee).

3. **Transporter Payment:** The system automatically deducts a **5% Platform Commission** from the delivery fee and transfers the remaining 95% to the Driver's account.

 Action (Automated Receipt): The system sends an automated notification to the Farmer: "Delivery Successful - Funds Transferred." The Farmer does not need to manually confirm anything.

 System Status: Completed / Ready for Rating

Step 5.3: Dispute & Return Process

 Action (If Rejected BEFORE Loading - from Step 4.2): Because the crops were poor quality before the journey started, the fault lies completely with the Farmer.

- **Buyer:** Receives a full 100% refund of their advance payment directly from the system, **including a full refund of the 3% Convenience Fee**.
- **Transporter:** Receives an "Empty Trip Fee" to cover their wasted travel time and fuel.
- **Farmer (The Penalty):** The "Empty Trip Fee" is automatically deducted from the Farmer's account. If the Farmer's account does not have enough money, it goes into a **negative balance** and is temporarily suspended. They cannot list new crops until this debt is paid back to the system.

 Action (If Rejected AFTER Delivery - from Step 5.1): Because the crops matched the photos at loading but arrived damaged, the fault lies completely with the Transporter.

- **Buyer:** Receives a full 100% refund. The delivery fee and the **3% Convenience Fee** they paid are fully refunded by the system. The money they paid for the crops is automatically deducted from the Transporter's account and given back to the Buyer.
- **Farmer:** Keeps the original advance payment they received for the crops as compensation for their ruined harvest.
- **Transporter (The Penalty):** Takes full responsibility for the damaged goods. They do not get paid for the delivery, and their account is charged to refund the Buyer for the ruined crops. If the Transporter's account does not have enough money, it goes into

a **negative balance** and is temporarily locked until they pay the system back. They earn zero money for the trip.

 System Status: Return Journey / Disputed

Step 5.4: Order Completion & Rating

- **Action (Manual User Rating):** Upon completion, the Buyer, Farmer, and Transporter can manually rate each other (1 to 5 Stars) based on their personal experience, such as communication, punctuality, and crop quality.
- **Action (Automated System Rating):** In the background, the system automatically adjusts each user's overall Trust Score based on the transaction data. Users gain positive score points for successful, dispute-free deliveries and lose points for canceled orders, delayed pickups, or causing refunds.
- **System Status:** Closed